

A Literature-Implied Affirmative Answer for Discrete Metric Spaces

Math discovery packet

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1 Status

This is a lightweight *literature-implied answer*, not an original new theorem. The source paper is

D. Bate and S. Li, *Thick Families of Geodesics and Differentiation*, arXiv:1904.02315.

In Section 7, after stating that the questions use “metric space” to mean “complete metric space,” it asks:

Does every discrete metric space biLipschitz embed into some RNP space?

The supporting paper is F. Albiac, C. Gartland, A. Petitjean, and A. Prochazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, arXiv:2103.09370.

2 Supporting theorem

ArXiv:2103.09370 proves the following characterization for Lipschitz-free spaces. For any metric space M , the completion of M is purely 1-unrectifiable if and only if the Lipschitz-free space $\mathcal{F}(M)$ has the Radon-Nikodým property. The same paper records the corresponding embedding consequence: a complete purely 1-unrectifiable metric space embeds isometrically into a Banach space with the Radon-Nikodým property.

Recall also the canonical metric embedding $x \mapsto \delta_x$ of M into $\mathcal{F}(M)$.

3 Identification

Let M be a complete discrete metric space. We show that M is purely 1-unrectifiable. Let $A \subset \mathbb{R}$ and let $f : A \rightarrow M$ be Lipschitz. Since A is separable and f is continuous, $f(A)$ is separable.

A separable subspace of a discrete metric space is countable: if D is a countable dense set in a discrete subspace Y , then every singleton $\{y\}$ is open in Y , so density forces $D \cap \{y\} \neq \emptyset$ for each $y \in Y$. Hence $Y = D$.

Thus $f(A)$ is countable, and therefore $\mathcal{H}^1(f(A)) = 0$. This proves that M is purely 1-unrectifiable. Because M is complete, the theorem of Albiac–Gartland–Petitjean–Prochazka applies directly: $\mathcal{F}(M)$ has the RNP and M embeds isometrically into $\mathcal{F}(M)$.

4 Conclusion

Under the source paper's stated convention that the questions concern complete metric spaces, Question (Q4) has an affirmative answer in later literature. Indeed, every complete discrete metric space embeds not merely bi-Lipschitzly, but isometrically, into a Banach space with the Radon-Nikodým property.

5 Scope

This packet only addresses Q4. It does not settle the characterization questions Q1–Q3 or the universal-distortion question Q5. The supporting paper does not appear, from the local cheap-index pass, to explicitly cite the source Q4; the answer recorded here is the direct implication above.

6 References

- D. Bate and S. Li, *Thick Families of Geodesics and Differentiation*, arXiv:1904.02315.
- F. Albiac, C. Gartland, A. Petitjean, and A. Prochazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, arXiv:2103.09370.